CODEX PLATFORM - QA APPLICATION TEST PLAN

Comprehensive Test Suite for Super Admin, Certificates & Session Security

Target Roles: Super Admin / Administrator | Environment: Staging & Production

MODULE 1: TWO-TIER CERTIFICATE APPROVAL WORKFLOW

Bulk Certificates

TC-CERT-01: Regular Admin Batch Submission (Pending Approval)

Admin

Steps: Log in as regular Admin -> Bulk Certificates -> Fill event name, select template, upload CSV -> Click 'Submit Certificates Batch'.

Expected: Batch is created in PENDING_APPROVAL status. No certificates are generated yet. Success toast indicates submission is queued for Super Admin review.

TC-CERT-02: Direct Super Admin Authorization & Issuance

Super Admin

Steps: Log in as Super Admin -> Bulk Certificates -> Fill details & upload student list -> Click 'Authorize & Generate Immediately'.

Expected: Direct generation triggers immediately. Batch completes and dispatched certificates are sent via email.

TC-CERT-03: Super Admin Batch Review & Approval

Super Admin

Steps: Navigate to 'Approval Requests & Batches' tab -> Click pending batch -> Inspect student roster -> Click 'Approve Batch'.

Expected: Batch status transitions to PROCESSING then COMPLETED. Certificates generated cryptographically and dispatched to all recipient emails.

TC-CERT-04: Super Admin Batch Rejection with Notes

Super Admin

Steps: Navigate to pending batch in 'Approval Requests & Batches' -> Click 'Reject Batch' -> Enter rejection reason -> Confirm.

Expected: Batch transitions to REJECTED. Submitting admin sees reason in batches view. No certificates generated.

MODULE 2: PRE-GENERATION BATCH & ROSTER EDITING

Metadata & Recipient Roster

TC-EDIT-01: Edit Batch While in PENDING_APPROVAL Status

Admin / Super

Steps: Under Batches tab, locate a PENDING_APPROVAL batch -> Click 'Edit Batch' button -> Open Edit Modal.

Expected: Modal loads current event name, date, type, and full student roster. All fields are editable.

TC-EDIT-02: Update Student Roster Prior to Generation

Admin / Super

Steps: In Edit Modal, add a new student row, delete a mistyped recipient, or re-upload replacement CSV -> Click 'Save Changes'.

Expected: Roster saves successfully. Student count updates in table. New recipient list will be used upon generation.

TC-EDIT-03: Locked State Enforcement (Negative Test)

Admin / Super

Steps: Inspect batches in APPROVED, PROCESSING, COMPLETED, or REJECTED status -> Verify 'Edit Batch' is replaced by locked badge.

Expected: Edit button is absent. Direct API call PUT /api/v1/certificates/batches/:id returns 400 Bad Request error ('Only batches in PENDING_APPROVAL status can be edited').

MODULE 3: SESSION GOVERNANCE & REVOCATION

Security & Sessions

TC-SESS-01: Regular Admin Session Isolation

Admin

Steps: Log in as regular Admin -> Open /admin/sessions -> Check displayed active sessions.

Expected: Only sessions belonging to the logged-in administrator are shown. Sessions of other admins or Super Admin are completely hidden.

TC-SESS-02: Super Admin Global Session Audit

Super Admin

Steps: Log in as Super Admin -> Open /admin/sessions -> Check displayed sessions list.

Expected: Super Admin can see all active and historical sessions across all admins. Each session displays admin name and email.

TC-SESS-03: Revocation Permission & Execution

Super Admin

Steps: Find an active session of a secondary admin -> Click 'Revoke Access' -> Confirm modal.

Expected: Session status changes to REVOKED immediately. The affected administrator is logged out on their next request.

TC-SESS-04: Timestamp Sanity & Legacy Resiliency

All Roles

Steps: Inspect session creation & expiry times across devices -> Verify date formatting.

Expected: All timestamps display formatted dates (e.g., 'Oct 14, 11:47 PM'). No 'Invalid Date' errors appear.

MODULE 4: ADMIN MANAGEMENT ISOLATION

Admin Accounts (/admin/admins)

TC-ADM-01: Super Admin Exclusion from Admin List

Super Admin

Steps: Navigate to /admin/admins -> Inspect table and search counters.

Expected: The root Super Admin account is NOT listed in the table. The list exclusively displays secondary provisioned staff admins.

TC-ADM-02: Provisioning & Actions on Secondary Admins

Super Admin

Steps: Click 'New Admin' -> Provision account -> Edit credentials -> Test Delete Admin.

Expected: All provisioned admins can be created, edited, and deleted cleanly. Total Administrators counter matches provisioned admins only.

TC-ADM-03: Root Email Collison Prevention (Negative Test)

Super Admin

Steps: Try creating a new admin using the Super Admin .env email address.

Expected: Action is rejected with 400 Bad Request ('Cannot create an admin using the Super Admin email address').

MODULE 5: EXCLUSIVE 'ROOT CONTROL • SUPER ADMIN' TAG

UI / UX Branding

TC-TAG-01: Super Admin Exclusive Visual Tag Presence

Super Admin

Steps: Log in as Super Admin -> Inspect: Sidebar, Mobile Header, Profile, Sessions, Dashboard Header, Welcome Card, Logs, and Bulk Certificates.

Expected: The glowing 'ROOT CONTROL • SUPER ADMIN' tag with pulsing indicator appears prominently across all specified locations.

TC-TAG-02: Regular Admin Tag Absence (Negative Test)

Admin

Steps: Log in as Regular Admin -> Inspect same locations across the suite.

Expected: 'ROOT CONTROL • SUPER ADMIN' tag is NEVER visible. Only standard 'Admin', 'Admin Portal', or 'Administrator' badges are rendered.

MODULE 6: AUDIT LOGGING & ACTIVITY ATTRIBUTION

System Logs (/admin/logs)

TC-LOG-01: Root Tag in Logs Header and Action Bar

Super Admin

Steps: Open /admin/logs -> Inspect top header badge and activity logs controls bar.

Expected: Header badge displays 'ROOT CONTROL • SUPER ADMIN'. Clear Logs button is active for Super Admin.

TC-LOG-02: Super Admin Action Attribution

Super Admin

Steps: Perform an action (approve batch, revoke session) -> Check Activity Logs table.

Expected: Log entry is recorded with adminId populated. Super Admin row has 'ROOT CONTROL • SUPER ADMIN' tag next to the admin name.